

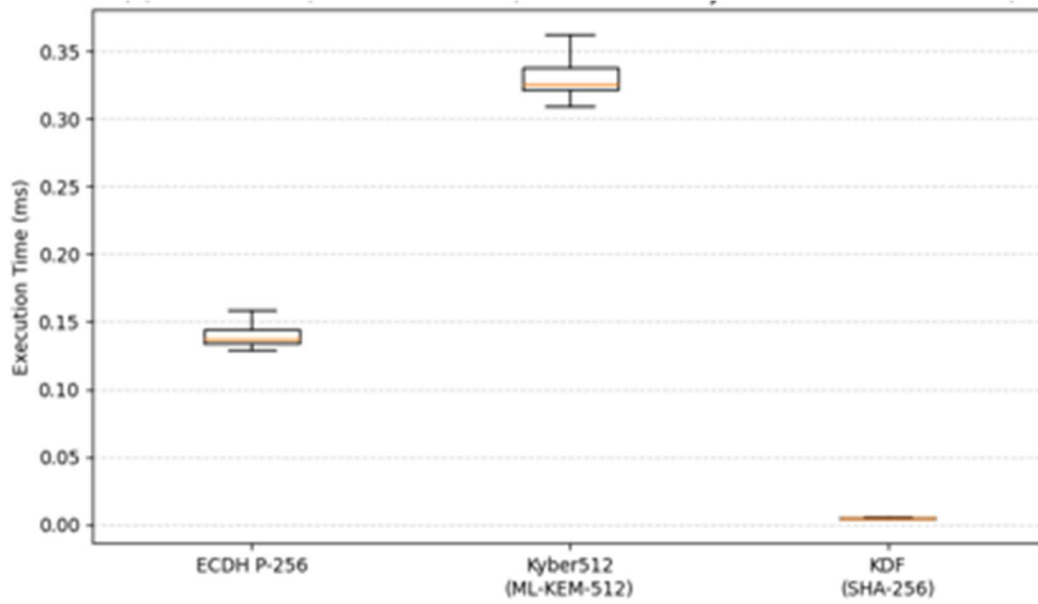


Figure V.6: Box plot comparing the latency distribution (ms) of each component in the hybrid ECDH + ML-KEM-512 key establishment workflow. The figure shows that ML-KEM-512 and ECDH P-256 account for the majority of total hybrid latency, while the SHA-256 KDF contributes less than 1% of the total overhead

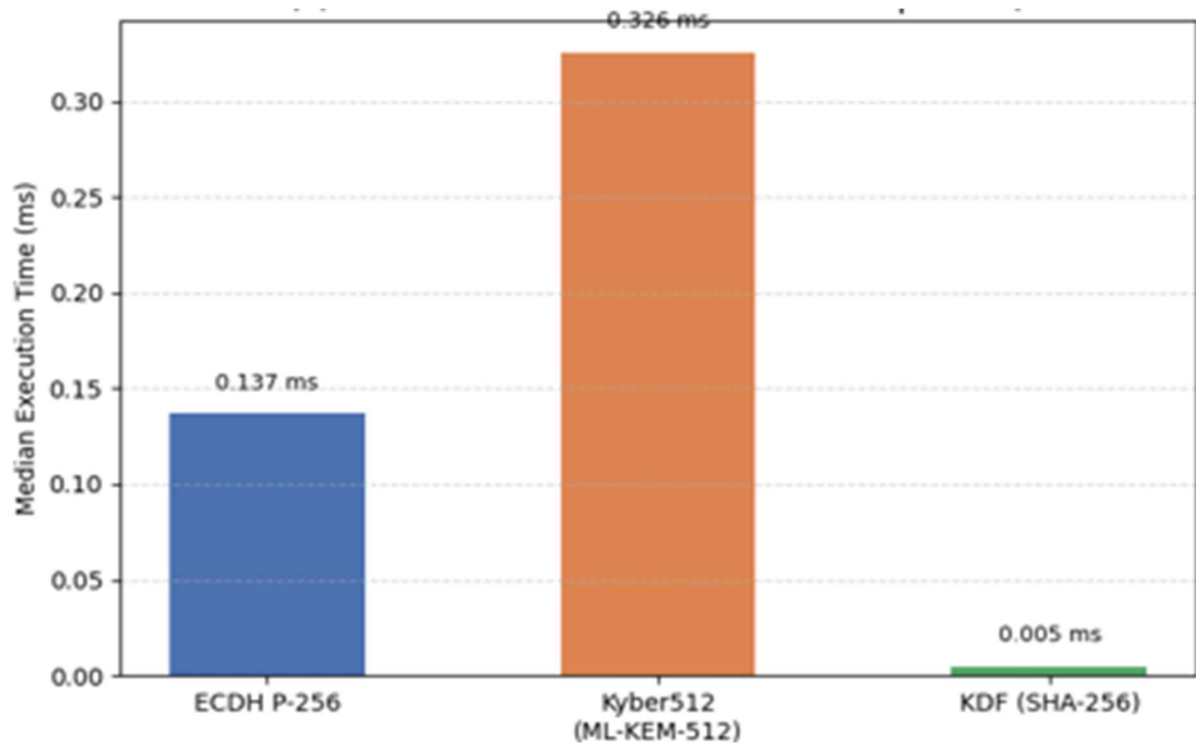


Figure V.7: Median execution time (ms) for each component of the hybrid ECDH + ML-KEM-512 workflow. ML-KEM-512 and ECDH P-256 account for the dominant share of total latency, while the SHA-256 KDF contributes less than 1% of the total hybrid overhead